Composite functions

1

a $f(-2) = -32$

b $g(f(-2)) = -27$

c $f(g(0)) = 500$

2 $f(g(2)) = 8$

3

a $f(7) = -20$

b $g(f(7)) = -107$

c $g(7) = 28$

d $f(g(7)) = -62$

e No

4

a $f(g(x)) = (9x-4)^3$

b $f(g(x)) = \left(x^2+1\right)^2$

c $f(g(x)) = \sqrt{4x-3}$

d $f(g(x)) = 4x^3+9$

e $f(g(x)) = 4x^2-3$

5

a $g(6) = 136$

b $f(g(6)) = -270$

c $r(f(g(6))) = 802$

6

a $g(f(6)) = 1596$

b $h(x) = -8x^2$

c Quadratic

7

a $r(x) = -2x^2+6$

b $q(x) = -6x^2+19$

8

a $f(2x) = 4x^2+3$

b No solution provided.

9

a Quadratic function

b $g(x) = 5x^2$

c $g(f(x)) = 45x^2-600x^3+2000x^4$

10

a $r(x) = x^2+6x+8$

b No solution provided.

c $p(q(x)) = x^2+2$

d $x < -1$

11

a $f(x)\,g(x) = 4x + \dfrac{12}{x}$

b $f(g(x)) = \dfrac{4}{x^2+3}$

c $\dfrac{f(x)}{g(x)} = \dfrac{4}{x(x^2+3)}$

d $\dfrac{g(x)}{f(x)} = \dfrac{x\left(x^2+3\right)}{4}$

12

a $f(x)\,g(x) = (9x+4)(x+1)(x-3)$

b $f(g(x)) = 9x^2 - 18x - 23$

c $\dfrac{f(x)}{g(x)} = \dfrac{9x+4}{x^2-2x-3}$

d $\dfrac{g(x)}{f(x)} = \dfrac{x^2-2x-3}{9x+4}$

Domain and range

13

a

i $f(g(x)) = \dfrac{9}{1-\dfrac{6}{x}}$

ii $(-\infty, 0) \cup (0, 6) \cup (6, \infty)$

iii $g(f(x)) = \dfrac{1-2x}{3}$

iv $\left(-\infty, \dfrac{1}{2}\right) \cup \left(\dfrac{1}{2}, \infty\right)$

b

i $f(g(x)) = |x|$

ii $(-\infty, \infty)$

iii $g(f(x)) = x$

iv $[-4, \infty)$

14 $f(x) = \sqrt{1+x}$

15

a $f(x) = \dfrac{1}{x^6}$

b $g(x) = (x-2)^6$

16 $g(x) = x$

17

a

i $x \in (-\infty, \infty), y \in [0, \infty)$

ii $x \in (-\infty, \infty), y \in (-\infty, \infty)$

iii $x \in (-\infty, \infty), y \in [0, \infty)$

b

i $x \in [0, \infty), y \in [0, \infty)$

ii $x \in (-\infty, \infty), y \in (-\infty, \infty)$

iii $x \in [3, \infty), y \in [0, \infty)$

c

i $x \in [0, \infty), y \in [0, \infty)$

ii $x \in (-\infty, \infty), y \in (-\infty, \infty)$

iii $x \in \left[-\frac{9}{4}, \infty\right), y \in [0, \infty)$

d

i $x \in (-\infty, \infty), y \in (-\infty, \infty)$

ii $x \in (-\infty, \infty), y \in (-\infty, \infty)$

iii $x \in (-\infty, \infty), y \in (-\infty, \infty)$

18

a $f(g(x)) = \sqrt{5x - 1}$

b $x \geq \frac{1}{5}$

c $g(f(x)) = 5\sqrt{x + 5} - 6$

d $x \geq -5$

19

a $f(g(x)) = -\frac{2x}{x - 5}$

b $x = 5$

c $g(f(x)) = \frac{2x}{2x + 5}$

d $x = -\frac{5}{2}$

Applications

20

a $g(x) = \frac{3}{4}x$ or $g(x) = 0.75x$

b $f(x) = x - 5$

c $f(g(x)) = \frac{3}{4}x - 5$ or $f(g(x)) = 0.75x - 5$

d $\$65.50$

21

a $A(m) = 1.6m + 17.6$

b $\$53.60$